AeroSense assembly and use feedback

Participant code: _____ Date: _____ Manual version: _____ Observer code: _____

This session identifies problems in the design and instructions; it is not a test of your ability. Tell us when a step is unclear, and stop whenever you wish. Use dry, cell-free parts with power off for assembly. UI mode: □ DEMO □ Previously verified hardware.

Data use: □ I agree that anonymous task times, errors and written feedback may be summarized in the team's iGEM project documentation.

Separate photo/video choice: □ Hands/parts may be photographed or video-recorded, without my face □ No photographs or video. Declining photographs/video does not prevent participation. Do not enter a name, telephone number or student ID.

Previous experience

Assembling small instruments/3D-printed parts: □ None □ Once or twice □ Regularly.

ESP32 or similar electronics: □ None □ Some □ Familiar.

Involved in this design: □ No □ Yes. Session: □ First attempt □ Retest after manual revision.

Observed tasks (observer completes)

Supply the same manual, parts and tools without a spoken demonstration. Time includes reading. Record every prompt. Stop possible damage or incorrect power connection and record why. Choose independent completion, assisted completion or incomplete; do not reduce all outcomes to success/failure.

Task	Seconds	Outcome	Prompts	Where the participant became stuck or made an error
1 Find the print files, materials and purchased items				
2 Identify fan, PCB and well orientations and fixing locations				
3 Position seals, plate and lid without forcing the assembly				
4 Identify power/USB, switch, programming interface and well replacement				
5 Locate stop, calibration, status and data download in the UI				

Possible damage, omitted critical seal, incorrect power connection or confusion between DEMO and measurements? □ No □ Yes: _____

Relevant manual page/figure and participant's exact words: _____

Your ratings

1 strongly disagree; 2 disagree; 3 neutral/uncertain; 4 agree; 5 strongly agree. Select not applicable for unperformed tasks. This is a project-specific questionnaire, not SUS.

Statement	1	2	3	4	5	N/A
I could find the correct parts and print files.	□	□	□	□	□	□
The figures made part orientation clear.	□	□	□	□	□	□
I knew which screw or seal belonged in each step.	□	□	□	□	□	□
I could assemble the parts without forcing them.	□	□	□	□	□	□
I knew how to replace a well without touching the optical components.	□	□	□	□	□	□
I could distinguish DEMO from actual hardware readings.	□	□	□	□	□	□
I knew when to stop rather than continue measuring.	□	□	□	□	□	□
I could distinguish the illuminated blank, fluorescence phase reference and living-cell F0 baseline.	□	□	□	□	□	□

The least clear step was: _____

A figure, phrase or part label that should change: _____

If only one thing could be improved, what would you choose? _____

Where did you worry about breakage, leakage or a wrong operation? _____

What help would you need to do this independently next time? _____

Team action and retest (after the participant leaves)

Issue ID: _____ Severity: □ Potential damage/wiring/leak □ Prevents completion □ Needs help □ Inconvenient.

Evidence (task, quotation, photo): _____

What actually changed: _____

Changed file/page/date: _____

Retest participant code: _____ Same person? □ Yes □ No. Same-task retest time: _____ s.

Retest outcome and remaining problem: _____

Publish individual times/issues, optionally medians. Explain small-sample and retest learning effects. Leave unperformed tests blank; do not fill in favorable responses.